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MARCIA™ ECONOMICS PAPER

The Economics of Long-Term Video Retention

Disk vs cloud vs MARCIA: a reproducible three-, five- and ten-year cost model for 500 TB to 5 PB of surveillance video



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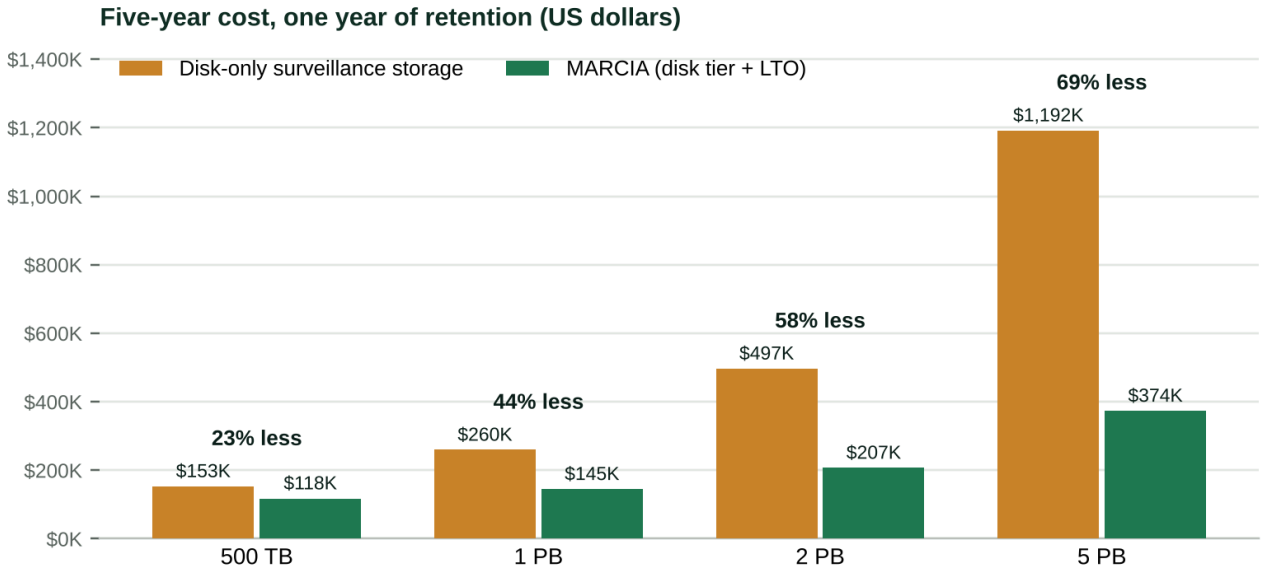
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Summary

Most surveillance storage decisions are made on purchase price. Most of the money is spent afterwards: power and cooling around the clock, replacement drives, and above all the cost of keeping more video every year as camera counts, resolutions and retention requirements grow.

This paper compares three ways to keep one year of video from a surveillance system:

- **Disk-only:** a complete surveillance storage system with enough RAID 6 disk to hold the whole retention period.
- **Cloud:** the video stored with cloud storage providers at their published standard prices.
- **MARCIA:** a small disk tier (10% of the video) for recent footage, with everything else on LTO tape, managed by Cozaint MARCIA™ Intelligent Video Lifecycle Management.



What the numbers show. At 500 TB, MARCIA costs **23% less** than disk-only over five years. The advantage grows with scale: **44% less at 1 PB, 58% at 2 PB and 69% at 5 PB.** Against cloud storage, MARCIA costs **77% to 92% less**. The more video an organization keeps, the more the lifecycle approach saves.

Every figure in this paper comes from the same price model that powers the storage cost calculator at **cozaint.com/assessment**, using prices checked on November 1, 2026. Anyone can reproduce the results there, change the assumptions and see the line-by-line math.

The results

Five years of ownership

Archive size	MARCIA (disk tier + LTO)	Disk-only system	Cloud storage	MARCIA saves vs disk	MARCIA saves vs cloud
500 TB	$117,762	$153,229	$508,680	23%	77%
1 PB	$145,062	$259,628	$1,003,080	44%	86%
2 PB	$207,050	$496,739	$1,991,880	58%	90%
5 PB	$373,530	$1,191,981	$4,958,280	69%	92%

Three years of ownership

Archive size	MARCIA (disk tier + LTO)	Disk-only system	Cloud storage	MARCIA saves vs disk	MARCIA saves vs cloud
500 TB	$107,951	$150,067	$305,208	28%	65%
1 PB	$128,587	$254,598	$601,848	49%	79%
2 PB	$176,870	$488,709	$1,195,128	64%	85%
5 PB	$303,465	$1,174,414	$2,974,968	74%	90%

Ten years of ownership

Archive size	MARCIA (disk tier + LTO)	Disk-only system	Cloud storage	MARCIA saves vs disk	MARCIA saves vs cloud
500 TB	$151,793	$189,645	$1,017,360	20%	85%
1 PB	$195,753	$329,229	$2,006,160	41%	90%
2 PB	$301,507	$630,862	$3,983,760	52%	92%
5 PB	$577,995	$1,502,010	$9,916,560	62%	94%

At 500 TB the gap narrows over ten years because MARCIA's cartridges are erased and reused each year under the Media Lifecycle Renewal service, while the disk model assumes the same system simply keeps running. The model still adds replacement drives for the disk-only system at the end of its five-year rated life, but not a new chassis, servers or controllers, so the ten-year disk figure is conservative.

How to read these numbers

1. **Scale changes the answer.** The tape tier carries fixed costs (a library, drives, software), so it pays off as the archive grows. Below roughly 500 TB the two approaches come closer together; above 1 PB, MARCIA's advantage is large and grows.
2. **Retention changes the answer.** Disk cost grows in direct proportion to the amount of video kept. With MARCIA, keeping more video mostly means adding cartridges.
3. **Cloud is the most expensive way to keep a year of video.** Recurring monthly fees on every terabyte add up quickly, before counting the internet connection and retrieval fees.

Assumptions

Every assumption below is visible, and can be changed, in the calculator at cozaint.com/assessment.

Assumption	Value used
Retention	365 days: each day's video is kept for one year
LTO generation	LTO-9, 18 TB per cartridge (the calculator default; LTO-10 is also available at 30 TB, and up to 40 TB with the newest cartridges)
LTO drives	3 library drives (the calculator default)
MARCIA disk tier	10% of the video kept on disk for instant playback
Disk-only system price	$242.20 per TB of raw disk: the average of six complete video surveillance storage systems from verified current providers, at US reseller prices, with volume price breaks for larger systems (15% at 1 to 2 PB, 18% at 2 to 5 PB, 21% at 5 to 10 PB)
RAID	RAID 6 in 12-drive groups (10 data + 2 parity), so video needs about 20% more raw disk than its size
Hard drives	18 TB enterprise drives at $792; rated for 5 years of 24/7 use; failure rate from the manufacturers' published reliability ratings
Disk failures in the disk-only system	Assumed covered by the system warranty (no charge)
Electricity	13.87¢ per kWh, the US average commercial price for the 12 months ending August 2026 (EIA data)
Cooling	Power use multiplied by 1.52 (a typical data-room overhead)

Assumption	Value used
LTO cartridges	Cozaint Certified LTO media at list price; rated for 30 years in archive, so no replacements are counted
Cozaint products	MARCIA LTO library, drives, SAS connectivity, media, Media Lifecycle Renewal and MARCIA base software at Cozaint list price (MSRP)
Cloud storage	Average of four cloud providers' standard, pay-as-you-go storage prices at each archive size (about $16.50 to $17 per TB per month)

The comparison leans in the competitors' favor. Disk systems and cloud are priced at discounted reseller and published prices, while Cozaint is at full list price. Not counted for disk-only: rack and floor space, support contracts, and the full system replacement most organizations make after five years. Not counted for cloud: the on-site recorder the cameras still need, the internet connection to upload every camera around the clock, retrieval (download) fees and minimum-storage fees.

A worked example: 1 PB for five years

This is the full calculation behind one row of the five-year table: 1 PB of video (one year of retention), owned for five years.

MARCIA (disk tier + LTO)

Item	How it is worked out	Five-year cost
Cozaint Certified LTO cartridges	56 × $349.00 (18 TB each)	$19,544
MARCIA Media Lifecycle Renewal	224 renewals × $119.00. The 56 cartridges fill up after 365 days; after that each is erased and reused about 4 times over 5 years	$26,656
MARCIA LTO library	80-slot MARCIA LTO library (COZ-MAR-LIB80), 80 cartridge slots	$9,399
LTO library drives	3 × $16,149.00 (LTO-9 SAS library drive, COZ-MAR-DRV-09)	$48,447
SAS connectivity	1 × $199.00 adapter + 3 × $99.00 cables	$496
MARCIA base software	1 × $599.00	$599

Item	How it is worked out	Five-year cost
10% disk tier: hard drives	12 × $792.00 (18 TB drives; 100 TB, 10% of the video, kept on disk for instant playback; market price)	$9,504
10% disk tier: enclosure	1 × $5,800.74 (market price)	$5,801
10% disk tier: server	1 × $15,956.82 (runs MARCIA and the disk tier; market price)	$15,957
Replacement drives: failures	1 × $792.00 (disk tier only. Tape cartridges do not spin, so they do not wear out this way)	$792
Replacement drives: worn out	None within 5 years. Drives are rated for 5 years of 24/7 use; keep the video longer and every drive must be replaced	$0
Replacement cartridges	None. LTO cartridges are rated for 30 years in archive, far longer than this 5-year period	$0
Power and cooling	1.3 kW around the clock for 5 years, cooling included. Cartridges on the shelf use no power	$7,867
Total		**$145,062**

Disk-only surveillance system

Item	How it is worked out	Five-year cost
Complete surveillance storage system	1,000 TB of video needs 1,200 TB of raw disk (RAID 6) × $242.20 per TB, the average price of 6 complete video surveillance storage systems, less a volume price break of 15% for a system 1 PB to 2 PB ($205.87 per TB)	$247,052
Replacement drives: failures	Included: assumed covered by the system warranty	$0
Replacement drives: worn out	None within 5 years. Drives are rated for 5 years of 24/7 use; keep the video longer and every drive must be replaced	$0

Item	How it is worked out	Five-year cost
Power and cooling	About 2.1 kW around the clock for 5 years at 13.87¢ per kWh, cooling included	$12,577
Total		**$259,628**

Cloud storage

Item	How it is worked out	Five-year cost
Cloud storage	1,000 TB × $16.72 per TB per month (average of 4 cloud providers at this size) × 60 months	$1,003,080
Total		**$1,003,080**

What the model does not show, but customers feel

Power, cooling and space

A cartridge in a library slot or on a shelf draws no power and needs no cooling. Disk spins 24 hours a day, every day, and every watt of it must also be cooled. In the 1 PB example, the disk-only system draws about 2.1 kW around the clock against 1.3 kW for the whole MARCIA system, and that gap widens with every petabyte. Tape industry studies put tape's long-term energy use at about 87% below an equivalent amount of disk.

Replacement cycles

Enterprise drives are rated for about five years of continuous use. Keep video longer, or keep the system running longer, and every drive must be replaced. LTO cartridges are rated for decades in archive.

Growth

When cameras are added or retention is extended, a disk-only system needs more drives, more enclosures and eventually more servers. A MARCIA system mostly needs more cartridges, and a library that grows by expansion modules.

Risk

LTO cartridges can be exported and stored offline, out of reach of the network, and WORM cartridges can be used where policy requires recordings that cannot be overwritten. Disk-only systems keep all video online; cloud depends on the provider and the internet connection.

The value of the video itself

Storage savings start the conversation. The bigger reason to keep more video is what it is worth: evidence for investigations and litigation that surface months later, proof of compliance, and history that new AI analytics can re-examine. Cheaper retention means organizations no longer have to choose between their budget and their evidence.

Proof from a real deployment

At the City of Oceanside's San Luis Rey Wastewater Treatment Plant, more than 30 cameras recording 1080p video 24/7 needed one year of retention, about 548 TB. A two-tier disk-and-LTO design, the architecture MARCIA is built on, came in at **54% lower five-year cost** than disk-only, as calculated at the time with the FUJIFILM video surveillance retention cost calculator ($31,515 against $68,506).

> "It is always difficult to find a system that fits within our budget and meets our long-term retention requirements. Cozaint's askALICE has been an ideal fit for our surveillance recording needs and our budget."
>
> Scott Prather, Senior IT Analyst, City of Oceanside, California

Run your own numbers

The figures here use typical assumptions. Your result depends on your cameras, resolution, frame rate, motion, retention and how long you will own the system.

1. Go to **cozaint.com/assessment**.
2. Enter your camera count, resolution, frame rate and retention.
3. Choose the LTO generation and adjust any hard drive assumption you want to test.
4. Download the PDF report with the full cost breakdown.

Want an engineer to check the design? Request a MARCIA assessment at **cozaint.com/marcia**, or contact **info@cozaint.com** · **760-975-8000**. Cozaint Corporation, headquarters Palmetto Bay, Florida, office San Diego, California. Prices checked November 1, 2026; results are estimates for comparison, not quotations.